

Business Analysis — Research Assignment

FreshMart Supermarkets:

Why Is Our Basket Getting Smaller?

A data-driven investigation into declining transaction value across a national grocery chain.

Module	Business Analysis & Data-Driven Decision Making
Assignment type	Individual / small-group research assignment
Weighting	40% of module mark
Total marks	100
Duration	3 weeks
Deliverables	Written report + presentation deck + supporting analysis

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1 Assignment overview

In this assignment you step into the role of a Business Analyst investigating a real-shaped commercial problem for a fictional grocery retailer, FreshMart Supermarkets. You are given a business situation that is deliberately *under-specified* — exactly as it would arrive on a real analyst's desk — together with a dataset of transactions, stores and operational events. Your job is to turn a vague executive concern into a defined problem, investigate it with evidence, and present recommendations that leadership can act on.

This is a **research assignment**, not a set of pre-defined exercises with one right answer. The marks reward how well you frame the problem, how rigorously you analyse the data, how deep your insight goes, and how clearly you communicate — the core competencies of a working Business Analyst.

The one-line brief

“Footfall is up, but our sales aren’t growing. It feels like people are buying less per visit. Find out what is going on, and tell us what to do about it.”

— Head of Retail Operations, FreshMart Supermarkets

2 The scenario

FreshMart is a mid-sized grocery chain operating 42 stores across three South African provinces (Gauteng, Western Cape and KwaZulu-Natal), in three store formats: Large, Medium and Express.

Over the past year, leadership has become uneasy. The signals appear contradictory:

- **Footfall is up.** More customers are walking through the doors than a year ago (the business reports roughly +6% year-on-year).
- **Revenue is not growing to match.** Total sales have stayed broadly flat — and by some measures have slipped slightly.
- **Baskets feel smaller.** Store managers sense that customers are spending less per visit, but nobody can say by how much, where, or why.

The Head of Retail Operations has a hunch — “people are buying less per visit” — but a hunch is not a plan. She needs an analyst to establish *whether* the effect is real, *where* it is concentrated, *what* is driving it, and *what interventions* would recover the lost value — before the next budget cycle.

3 Learning outcomes

On successful completion of this assignment you will be able to:

- Translate an ambiguous business concern into a clearly scoped, measurable problem statement.
- Identify stakeholders and elicit and structure competing, testable hypotheses.
- Assess data availability and quality, and specify the data required to answer a business question.
- Apply appropriate analytical techniques — trend analysis, segmentation, cohort and comparison analysis — to isolate root causes.
- Distinguish correlation from a credible causal story, and avoid common analytical traps (for example mix-shift and Simpson’s paradox).
- Produce prioritised, actionable, cost-aware recommendations with defined success metrics.
- Communicate findings persuasively to a non-technical executive audience in writing and in a presentation.

4 Your role and the business problem

Your role

You are the Business Analyst assigned to the investigation. You report findings to a steering group made up of the Head of Retail Operations, the Merchandising Lead, the Pricing Manager and a sample of Store Managers. They are commercially literate but not technical — they want clarity, evidence and a decision, not code.

The problem, as you must sharpen it

The concern as stated is too vague to act on. Part of your task is to make it precise. Before you can answer “why is basket size falling?” you must decide, and justify, questions such as:

- Is “basket size” measured by rand value, by number of items, or both? (They can move in different directions.)
- Is the decline real and chain-wide, or concentrated in particular stores, formats, regions, customer groups or times?
- If footfall is up but revenue is flat, what does that arithmetic actually imply about basket value and trip frequency?

A strong submission treats the vague brief as the first thing to be *interrogated*, not accepted at face value.

5 The dataset

You are given a representative extract of FreshMart data for the 2025 calendar year. It is provided as an Excel workbook (FreshMart_Dataset.xlsx) and as individual CSV files. The extract contains three linked tables:

Table	Grain	What it contains
Transactions	One row per basket	~64,000 till transactions: store, date, day of week, loyalty flag, customer ID, item count, basket value (ZAR) and payment method.
Stores	One row per store	All 42 stores: province, format, floor area, year opened, region manager, and whether a discount competitor opened nearby (with the opening date).
Stockouts	One row per event	Recorded stockout events: store, date, category affected and approximate duration.

The tables join on **store_id**. A full field-by-field data dictionary is in Appendix A and on the *Data_Dictionary* sheet of the workbook.

Important notes about the data

- This is a sample of transactions, not every sale — use it to compare and find patterns, not to report FreshMart’s exact total turnover.
- A blank customer_id means the shopper was not a loyalty member. Treat blanks deliberately.
- Like real operational data, it is not perfectly clean. Checking and documenting data quality is part of the task, not a distraction from it.

6 Your research tasks

Work through the following six tasks. They mirror the workflow of a real BA engagement and map directly to the marking rubric in Section 10. You do not have to present them as six separate chapters, but your report and presentation should evidence all of them.

Task 1 — Define and scope the problem

- Restate the business concern as one or more precise, measurable problem statements.
- Define your metrics explicitly (for example how you measure basket size, and over what period).
- Confirm from the data whether the effect is real, and quantify it. State the size of the decline you find.

Task 2 — Stakeholders and hypotheses

- Identify the key stakeholders and what each cares about.
- Develop at least four competing, testable hypotheses for why basket size might be falling (for example: pricing, competition, product availability, changing shopping behaviour, store operations).
- For each hypothesis, state what evidence in the data would support or reject it.

Task 3 — Data preparation and quality

- Profile the data: row counts, date coverage, distributions, missing or anomalous values.
- Document any data-quality issues and how you handled them.
- Prepare any derived fields you need (for example month, weekday/weekend, basket-size bands, loyalty vs non-loyalty).

Task 4 — Analyse and segment

- Test your hypotheses. Move beyond the chain-wide average — the interesting story is almost always inside a segment.
- Segment and compare across dimensions such as time, store, format, province, loyalty status, basket type and shopping day.
- Where relevant, join the Stores and Stockouts tables to test store-level and operational hypotheses.

Task 5 — Findings and root cause

- Synthesise your analysis into a clear, evidence-backed account of what is driving the decline.
- Be explicit about which hypotheses the data supports, which it rejects, and how confident you are.
- Guard against being fooled by shifts in the mix of customers or trips rather than genuine per-customer change.

Task 6 — Recommendations, metrics and monitoring

- Recommend a prioritised set of interventions that follow logically from your findings.
- For each, note the expected impact, rough effort/cost, owner and any risks.
- Define how FreshMart will measure success (target metrics, and how you would set up a fair test — for example control vs treatment stores) and what an ongoing monitoring dashboard should track.

7 Guidance and suggested approach

How to think about it

Start with the arithmetic of the paradox. If more people are visiting but revenue is flat, then either they are visiting to buy *less each time*, or the *kind* of visit is changing. Both are measurable in this data. Let that logic guide where you look.

Then resist the chain-wide average. An average that falls a little can hide one group falling a lot while another holds steady. Always ask: falling for whom, where, and since when? The moment you split the data the right way, the story usually appears.

Techniques you might use

- **Trend analysis** — plot volume, average basket value and total revenue by month; look at the first half vs the second half of the year.
- **Segmentation** — compare averages across store format, province, loyalty status and basket-size bands.
- **Comparison / quasi-experiment** — compare stores that experienced an event against similar stores that did not, before vs after the event date.
- **Distribution analysis** — look at the shape of basket sizes, not just the mean. Is the average falling because everyone spends less, or because the mix of small vs large trips is changing?
- **Joining tables** — bring store attributes and operational events alongside transactions to test store-level and availability hypotheses.

Tools

Use whatever you are strongest in — Excel / pivot tables, SQL (BigQuery, PostgreSQL, etc.), Python (pandas), or Power BI / Looker Studio. Marks are for the quality of the analysis and reasoning, not the tool. Whatever you use, keep your working so it is reproducible.

What a strong submission looks like

- Frames the problem crisply before touching a chart; defines metrics; states assumptions.
- Goes beyond one headline number to a segmented, root-cause story supported by clear visuals.
- Names which hypotheses were rejected, not only the one that survived — that is where analytical maturity shows.
- Ends with a short, prioritised, costed set of actions and a way to measure whether they worked.

Common pitfalls to avoid

- Reporting only the chain-wide average and stopping there.
- Confusing a change in the mix of customers/trips with a change in individual behaviour.
- Claiming causation from a single correlation without a before/after or like-for-like comparison.
- Presenting twenty charts with no narrative. Executives want the story, then the evidence.
- Recommendations that don't follow from the evidence, or that can't be measured.

8 Deliverables

Submit three items:

1. **Written report (8–15 pages).** A structured business-analysis report: problem definition, approach, data and quality notes, findings with visuals, recommendations, success metrics, and any assumptions/limitations. Written for an executive reader — clear, concise, evidence-led.

2. **Presentation deck (10–15 slides).** A boardroom-ready deck telling the story: the problem, the key finding(s), the evidence, and the recommended actions. You will present this in 10 minutes followed by Q&A.
3. **Supporting analysis.** Your working files — SQL scripts, Python notebook, Power BI file or annotated Excel workbook — so your analysis is reproducible.

9 Presentation guidelines

Your presentation is assessed as a business communication, not a data dump. Aim for the flow an executive audience expects:

- **Situation & question** (1–2 slides) — the concern and the sharpened question.
- **Headline finding** (1 slide) — your answer, up front. Don't make them wait.
- **Evidence** (3–5 slides) — the segmentation and comparisons that prove it, one idea per slide.
- **Recommendations** (2–3 slides) — prioritised actions, expected impact, how you'll measure success.
- **Close** (1 slide) — the ask, and next steps.

Every chart should earn its place and carry a one-line “so what” takeaway as its title. Rehearse to time.
Storytelling and clarity carry real marks.

10 Marking rubric

Total: 100 marks.

Criterion	Marks	What earns the marks
Problem framing & scoping	10	Vague brief turned into precise, measurable problem statement(s); metrics defined; effect quantified.
Stakeholder analysis & hypotheses	10	Relevant stakeholders identified; ≥4 competing, testable hypotheses with expected evidence.
Data preparation & quality	10	Sound profiling; quality issues found, documented and handled; sensible derived fields.
Analysis rigour & technique	20	Correct, appropriate techniques applied and executed cleanly; reproducible working.
Depth of insight / root cause	15	Moves beyond averages to a segmented, credible root-cause story; hypotheses accepted and rejected.
Recommendations	15	Actionable, prioritised, cost-aware, and clearly following from the evidence.
Success metrics & monitoring	5	Sensible target metrics and a fair way to test and monitor impact.
Report quality	5	Clear structure, concise executive writing, effective visuals.
Presentation & storytelling	10	Compelling narrative, one idea per slide, well-timed, handles Q&A.
Total	100	

11 Suggested three-week timeline

Week	Focus
Week 1	Frame and scope the problem; stakeholder and hypothesis development; get to know the data; complete data-quality profiling. Deliver a one-page problem statement and hypothesis list.
Week 2	Core analysis and segmentation; test each hypothesis; isolate the root cause(s). Deliver draft findings with key charts.
Week 3	Turn findings into prioritised recommendations and success metrics; write the report; build and rehearse the deck. Submit and present.

12 Submission and academic integrity

- Submit the report (PDF), the deck (PDF or PPTX) and your supporting analysis files by the published deadline.
- You may use AI tools to assist your analysis and writing, but you must understand, be able to explain, and take responsibility for every claim you submit. Expect to justify your reasoning in Q&A.
- Cite any external sources. The analysis and conclusions must be your own work.
- There is no single “correct” answer. Marks reward rigour, insight and communication — a well-argued, evidence-based case is what we are looking for.

Appendix A — Data dictionary

Table	Field	Type	Description
Transactions	transaction_id	Text	Unique ID for each till transaction (basket).
Transactions	store_id	Text	Store where the transaction occurred → joins to Stores.
Transactions	transaction_date	Date	Calendar date of the transaction (2025).
Transactions	day_of_week	Text	Day name (Monday–Sunday).
Transactions	customer_id	Text	Loyalty customer ID; BLANK if not a loyalty member.
Transactions	is_loyalty_member	Text	Yes / No.
Transactions	num_items	Integer	Number of items in the basket.
Transactions	basket_value_zar	Decimal	Total basket value in ZAR.
Transactions	payment_method	Text	Card / Cash / Mobile.
Stores	store_id	Text	Unique store identifier (key).
Stores	store_name	Text	Store display name and location.
Stores	province	Text	Gauteng / Western Cape / KwaZulu-Natal.
Stores	store_format	Text	Large / Medium / Express.
Stores	floor_area_sqm	Integer	Retail floor area (m ²).
Stores	opened_year	Integer	Year the store opened.
Stores	region_manager	Text	Responsible regional manager.
Stores	has_nearby_competitor	Text	Yes / No — discounter opened within 2km in the period.
Stores	competitor_open_date	Date	Date the nearby discounter opened (BLANK if none).
Stockouts	stockout_id	Text	Unique ID for a stockout event.
Stockouts	store_id	Text	Affected store → joins to Stores.
Stockouts	date	Date	Date of the stockout.
Stockouts	category	Text	Product category out of stock.
Stockouts	duration_hours	Integer	Approx. hours the category was unavailable.
Stockouts	note	Text	Operational note.

Appendix B — Analysis starter checklist

Not a solution — a set of prompts to get you moving. You are not limited to these, and not every one will prove fruitful.

- What is the monthly trend in transaction volume, average basket value and total revenue across 2025?
- How does average basket value in the first half of the year compare with the second half?
- Split average basket value by loyalty status. Do members and non-members move the same way?
- Plot the distribution of basket sizes. Is it one hump or two? Is the mix of small vs large trips changing over time?
- Compare basket trends for stores with a nearby competitor against stores without one — before and after the competitor opened.
- Does store format (Large / Medium / Express) or province change the picture?
- Join the Stockouts table: on affected store-days, are baskets smaller? Does it matter which day of the week?
- Is there anything in payment method or day of week — or is that a red herring?
- For every pattern you find: is this a real behaviour change, or a shift in who is shopping / what kind of trip they are making?